# Comprehensive fixed-order analysis of $t\bar{t}W^{\pm}$ in the trilepton channel

Observable definitions and implementation reference

fNLO fixed-order template

30 August 2026

**Abstract**

This document specifies the 286-histogram HwU analysis `analysis_HwU_pp_ttxw_trilepton.o` for

$$pp \longrightarrow t\bar{t}W^{\pm} + X \longrightarrow b\,\bar{b}\;e\,e\,\mu\;3\nu + X.$$

The no-cut truth layer uses the electron and positron to analyse the top and antitop decays and the muon to analyse the associated $W$. The experiment-facing layer is formulated only in terms of the observable two-electron/one-muon final state, jets, and total missing transverse momentum. It includes both $e^+e^-\mu^{\pm}$ and $e^{\pm}e^{\pm}\mu^{\mp}$ charge patterns and does not assume which parent W produced a visible lepton. It includes the six distributions unfolded in the ATLAS differential measurement as well as extensive production, decay, polarization, spin-correlation, event-activity, transverse, and neutrino-blind reconstruction observables. One IR-safe measurement function is used for additive and multiplicative NLO decay-chain events and for every real-event counterevent.

## Contents

# 1 Scope and event interpretation

The flavour-specific theory interpretation is

$$
\begin{aligned}
t\bar{t}W^+ &\to b\,e^+\nu_e\;\bar{b}\,e^-\bar{\nu}_e\;\mu^+\nu_\mu,\\
t\bar{t}W^- &\to b\,e^+\nu_e\;\bar{b}\,e^-\bar{\nu}_e\;\mu^-\bar{\nu}_\mu.
\end{aligned}
$$

The event-level interface is more general: it requires exactly two status-one electron-flavour charged leptons, one muon, three light neutrinos, one $b$, and one $\bar{b}$. The charged-lepton sum must be $q_{3\ell} = \pm 1$, with two leptons of charge $q_{3\ell}$ and one of the opposite charge. Thus all charge/origin assignments of the $2e1\mu$ channel are accepted. Missing or duplicate required objects stop the run with a diagnostic.

At truth level the decay assignment is intentionally flavour-specific: $e^+\nu_e$ reconstructs the top-side $W^+$, $e^-\bar{\nu}_e$ the antitop-side $W^-$, and the unique muon and muon neutrino the associated $W^\pm$. These truth histograms are filled only when that exact PDG content is present. In particular, a $\mu^+$ (PDG $-13$) is paired with $\nu_\mu$ (PDG 14), and a $\mu^-$ (PDG 13) with $\bar{\nu}_\mu$ (PDG $-14$). Identification uses PDG codes rather than positions, so MadGraph process ordering is irrelevant.

The realistic layer does not consult ancestry or individual neutrino momenta. It sees two electron-flavour leptons, one muon, their charges, jets, and $\vec{p}_{\mathrm{T}}^{\,\mathrm{miss}}$. Consequently it remains defined when either a top decay or the associated W supplies the muon.

The analysis has three nested levels.

1. **Inclusive truth** has no acceptance cuts and uses exact neutrino momenta. It provides production and decay kinematics, perfect top/W reconstruction, polarization, and spin-density moments for the flavour-specific decay assignment above.

2. **Fiducial visible** uses only charged leptons, anti-$k_T$ jets, ideal partonic b tags, and total missing transverse momentum. It requires at least one accepted b jet and is the level closest to a particle-level measurement.

3. **Neutrino-blind reconstructed** additionally requires two distinct accepted b jets. A deterministic three-neutrino mass fit reconstructs all W bosons, both top quarks, and the full $t\bar{t}W$ system without reading a truth neutrino.

All active HwU weights are filled. Histogram integrals therefore retain the fixed-order cross-section normalization for the central, scale, PDF, and other configured weights; the module does not normalize shapes.

# 2 Kinematic conventions

Four-vectors are $p = (E, p_x, p_y, p_z)$, with

$$
p_{\mathrm{T}}(p) = \sqrt{p_x^2 + p_y^2}, \qquad y(p) = \frac{1}{2}\ln\frac{E + p_z}{E - p_z}, \tag{1}
$$

$$
\eta(p) = \frac{1}{2}\ln\frac{|\vec{p}| + p_z}{|\vec{p}| - p_z}, \qquad m(p) = \sqrt{\max(0, p^2)}. \tag{2}
$$

Azimuthal differences are wrapped into $[-\pi, \pi]$; unsigned $\Delta\phi$ lies in $[0, \pi]$, and $\Delta R = \sqrt{(\Delta\eta)^2 + (\Delta\phi)^2}$. Leading, subleading, and third refer to descending charged-lepton $p_{\mathrm{T}}$.

The associated-W charge is inferred without ancestry as $q_W = q_{3\ell} = \sum_{i=1}^{3} q_{\ell_i} = +1$ or $-1$. The unique same-sign pair comprises the two leptons with charge $q_W$; they are ordered by $p_{\mathrm{T}}$. The remaining lepton has charge $-q_W$. A "leading opposite-sign pair" combines this remaining lepton with the leading same-sign lepton. This convention is unambiguous also for $e^\pm e^\pm \mu^\mp$, where two opposite-sign $e\mu$ pairs exist. The dielectron pair is OSSF only when its charges are opposite. Multiplying an observable by $q_W$ permits direct charge-channel comparisons.

The missing transverse vector is computed from visible status-one particles,

$$
\vec{p}_{\mathrm{T}}^{\,\mathrm{miss}} = -\sum_{i \notin \{\nu_e, \nu_\mu, \nu_\tau\}} \vec{p}_{\mathrm{T},i}. \tag{3}
$$

For a consistent fixed-order point it equals the vector sum of all three neutrinos. Their difference is retained as a closure diagnostic, never as a selection.

For visible $v$ and massless transverse momentum $\vec{q}_{\mathrm{T}}$,

$$m_{\mathrm{T}}^2(v,\vec{q}_{\mathrm{T}}) = m_v^2 + 2\left(E_{\mathrm{T},v}|\vec{q}_{\mathrm{T}}| - \vec{p}_{\mathrm{T},v}\cdot\vec{q}_{\mathrm{T}}\right), \tag{4}$$

$$E_{\mathrm{T},v} = \sqrt{m_v^2 + p_{\mathrm{T}}^2(v)}. \tag{5}$$

The trilepton cluster mass uses $v = e_1 + e_2 + \mu$ and treats all missing momentum as one massless transverse object. The visible $m_{\mathrm{T2}}^{ee}$ is

$$\min_{\vec{q}_{1\mathrm{T}}+\vec{q}_{2\mathrm{T}}=\vec{p}_{\mathrm{T}}^{\,\mathrm{miss}}} \max\{m_{\mathrm{T}}(e_1,\vec{q}_{1\mathrm{T}}), m_{\mathrm{T}}(e_2,\vec{q}_{2\mathrm{T}})\}. \tag{6}$$

It is measurable but is not expected to have the clean dileptonic-$t\bar{t}$ endpoint because $\vec{p}_{\mathrm{T}}^{\,\mathrm{miss}}$ also contains the associated-W neutrino.

# 3 Truth resonances and decay radiation

The three exact W candidates are

$$p_{W_t^+} = p_{e^+} + p_{\nu_e}, \qquad p_{W_{\bar{t}}^-} = p_{e^-} + p_{\bar{\nu}_e}, \qquad p_{W_{\mathrm{assoc}}} = p_\mu + p_{\overset{(-)}{\nu_\mu}}.$$

At Born level $p_t = p_{W_t^+} + p_b$ and $p_{\bar{t}} = p_{W_{\bar{t}}^-} + p_{\bar{b}}$. For a real-emission event, each non-bottom status-one light quark or gluon is assigned to production, the top decay, or the antitop decay. The selected member of the $3^{N_{\mathrm{rad}}}$ possibilities minimizes

$$\chi^2_{\mathrm{truth}} = \left(\frac{m_{t,\mathrm{cand}} - m_t^{\mathrm{ref}}}{5\ \mathrm{GeV}}\right)^2 + \left(\frac{m_{\bar{t},\mathrm{cand}} - m_t^{\mathrm{ref}}}{5\ \mathrm{GeV}}\right)^2. \tag{7}$$

The associated W is not part of this QCD-radiation assignment. Reference masses are read from the model. This construction is independent of event record ordering and approaches the same measurement in every soft or collinear limit, apart from measure-zero assignment boundaries.

The top-side W helicity angles boost the charged lepton to its parent-W rest frame and use the W flight direction in the corresponding top rest frame. For the truth associated W, the helicity axis is its flight direction in the $t\bar{t}W$ rest frame and the analyser is the muon in its W rest frame. The reconstructed version uses whichever same-sign lepton the mass fit assigns to the associated W. Raw and $q_W$-signed angles are both plotted.

# 4 Jets and event activity

All status-one quarks with $|\mathrm{PDG}| \leq 5$ and gluons are clustered with FastJet using anti-$k_T$, $R = 0.4$, and four-vector recombination. Counted jets satisfy

$$p_{\mathrm{T}}(j) \geq 25\ \mathrm{GeV}, \qquad |\eta(j)| < 2.5,$$

matching the central particle-level jet definition of the ATLAS measurement. A jet is ideally b-tagged if it contains the unique final-state $b$ or $\bar{b}$ parton. If both bottom partons merge, inclusive and one-b fiducial plots remain defined, while the two-b reconstruction tier fails. Additional jets are accepted jets other than the two b-jet indices.

The activity variables are

$$H_{\mathrm{T}}^{j} = \sum_{\text{accepted jets}} p_{\mathrm{T}}(j), \qquad H_{\mathrm{T}}^{\ell} = p_{\mathrm{T}}(e_1) + p_{\mathrm{T}}(e_2) + p_{\mathrm{T}}(\mu), \tag{8}$$

$$H_{\mathrm{T}}^{\mathrm{extra}} = \sum_{j\notin b,\bar{b}} p_{\mathrm{T}}(j), \qquad H_{\mathrm{T}}^{\mathrm{vis}} = H_{\mathrm{T}}^{j} + H_{\mathrm{T}}^{\ell}, \tag{9}$$

$$S_{\mathrm{T}} = H_{\mathrm{T}}^{\mathrm{vis}} + p_{\mathrm{T}}^{\mathrm{miss}}, \qquad p_{\mathrm{vis}} = p_{e_1} + p_{e_2} + p_\mu + p_{b\mathrm{j}} + p_{\bar{b}\mathrm{j}}. \tag{10}$$

The analysis explicitly includes the six ATLAS unfolded variables: $N_{\mathrm{jets}}$, $H_{\mathrm{T}}^{j}$, $H_{\mathrm{T}}^{\ell}$, $\Delta R_{\ell b,\mathrm{lead}}$, $|\Delta\phi_{\ell\ell,\mathrm{SS}}|$, and $|\Delta\eta_{\ell\ell,\mathrm{SS}}|$.

# 5 Fiducial definition and cutflow

The fixed-order fiducial definition follows the central features of the ATLAS trilepton particle-level region:

- exactly two electron-flavour leptons and one muon with total charge $q_W = \pm 1$;
- the two same-sign leptons have $p_{\mathrm{T}} \geq 20$ GeV; the remaining lepton has $p_{\mathrm{T}} \geq 10$ GeV;

- $|\eta(e)| < 2.47$ and $|\eta(\mu)| < 2.5$;
- if an OSSF dielectron pair exists, $m_{ee} > 12$ GeV and $|m_{ee} - m_Z| > 10$ GeV; same-sign dielectron events pass these two conditions automatically;
- $|m_{3\ell} - m_Z| > 10$ GeV;
- at least one accepted jet and at least one accepted ideal b jet.

A parton-level isolation proxy requires every selected lepton and central accepted jet to satisfy

$$\Delta R(\ell, j) > \min\left(0.4,\ 0.04 + \frac{10\ \text{GeV}}{p_\text{T}(\ell)}\right). \tag{11}$$

This is intentionally stated as a proxy: exact experimental overlap removal acts sequentially on dressed leptons, particle jets, and b-hadron ghost tags, none of which exists in a pure partonic NLO QCD event. There is no missing-momentum cut in the fiducial definition.

The ten cutflow bin centres are:

| Bin | Accumulated requirement |
|---|---|
| 1 | valid visible $2e1\mu + 3\nu + b\bar{b}$ object content |
| 2 | three-lepton $p_\text{T}$ and $\eta$ acceptance |
| 3 | OSSF mass above 12 GeV when an OSSF pair exists |
| 4 | OSSF $Z$ veto when an OSSF pair exists |
| 5 | trilepton $Z$ veto |
| 6 | at least one accepted jet |
| 7 | at least one accepted b jet |
| 8 | lepton–jet separation; this defines the fiducial sample |
| 9 | two distinct accepted b jets; reconstruction input tier |
| 10 | successful three-neutrino reconstruction |

# 6 Three-neutrino reconstruction

Only the three charged leptons, their charges, two accepted b jets, and $\vec{p}_\text{T}^\text{miss}$ enter the reconstruction. A trial first assigns one of the two same-sign leptons to the associated W. The other same-sign lepton belongs to the same-charge top, while the unique opposite-sign lepton belongs to the other top. Both choices are tested. For each assignment, a trial chooses transverse momenta for the positive- and negative-top neutrinos; momentum conservation fixes

$$\vec{q}_{\text{assoc}\,\text{T}} = \vec{p}_\text{T}^\text{miss} - \vec{q}_{t^+\text{T}} - \vec{q}_{t^-\text{T}}.$$

Each equation $(p_\ell + p_\nu)^2 = (m_W^\text{ref})^2$ supplies up to two $p_z^\nu$ roots. A negative discriminant is replaced by its real part. All root combinations, both same-sign origin assignments, and both b-jet assignments are evaluated with

$$\chi^2_\text{reco} = \sum_{q=t,\bar{t}} \left(\frac{m_q - m_t^\text{ref}}{15\ \text{GeV}}\right)^2 + \sum_{V=W_t^+, W_{\bar{t}}^-, W_\text{assoc}} \left(\frac{m_V - m_W^\text{ref}}{10\ \text{GeV}}\right)^2 \tag{12}$$

$$+ \left(\frac{m_t - m_{\bar{t}}}{30\ \text{GeV}}\right)^2 + \left(\frac{\sum_i |p_{z,\nu_i}|}{1500\ \text{GeV}}\right)^2 + \left(\frac{\sum_i p_\text{T}(\nu_i)}{2000\ \text{GeV}}\right)^2. \tag{13}$$

The last two terms only resolve weakly constrained large-momentum solutions.

The four transverse degrees of freedom are initialized at equal sharing of the measured missing momentum. A $3^4$ coarse grid is followed by twelve four-coordinate pattern-search iterations. The procedure is deterministic, does not use truth matching, and returns all three W candidates, top, antitop, $t\bar{t}$, and $t\bar{t}W$. It is a transparent fixed-order reconstruction prescription rather than an emulation of a detector-specific likelihood fit.

# 7 Spin and polarization observables

The top-pair spin basis is constructed in the $t\bar{t}$ rest frame. With $\hat{p}$ the boosted positive-beam direction, $\hat{k}$ the top direction, and $c = \hat{p} \cdot \hat{k}$,

$$\hat{r} = \text{sign}(c)\frac{\hat{p} - c\hat{k}}{\sqrt{1-c^2}}, \qquad \hat{n} = \text{sign}(c)\frac{\hat{p} \times \hat{k}}{\sqrt{1-c^2}}. \tag{14}$$

A deterministic orthogonal completion covers forward exceptional points. The $e^+$ and $e^-$ are boosted to their parent rest frames and define six angles $c_a^+$, $c_b^-$, $a, b \in \{k, r, n\}$. All nine products $q_{ab} = c_a^+ c_b^-$ are recorded. For the convention

$$\frac{1}{\sigma}\frac{d^2\sigma}{dc_a^+ dc_b^-} = \frac{1}{4}[1 + B_a^+ c_a^+ + B_b^- c_b^- - C_{ab} c_a^+ c_b^-], \tag{15}$$

$B_a^\pm = 3\langle c_a^\pm\rangle$ and $C_{ab} = -9\langle q_{ab}\rangle$. The histograms themselves are raw moments and can be translated to another sign convention.

The associated-W suite adds the associated charged-lepton angle with the positive beam in the W rest frame, its charge-signed counterpart, its opening angle with the same-sign top-decay lepton, and

$$\mathcal{T}_{3\ell} = (\hat{\ell}_{t^+} \times \hat{\ell}_{t^-}) \cdot \hat{\ell}_{\text{assoc}}.$$

It also records $q_W \mathcal{T}_{3\ell}$ and, in the $t\bar{t}W$ frame, $q_W[\cos(\ell_{\text{assoc}}, t) - \cos(\ell_{\text{assoc}}, \bar{t})]$. At truth level $\ell_{\text{assoc}} = \mu$; the reconstructed tier uses the mass-fit assignment.

# 8 Complete histogram catalogue

Histogram numbers are stable implementation labels. Energies and momenta are in GeV; angles and rapidities are dimensionless.

## 8.1 Rates and production: histograms 1–34

| IDs | Histograms | Definition and purpose |
|---|---|---|
| 1 | inclusive rate | One-bin total valid $2e1\mu$ rate for either allowed charge pattern. |
| 2 | inclusive associated-W charge | $q_W = -1, +1$; charge ratio and relative asymmetry input. |
| 3 | cutflow | Ten cumulative stages defined above. |
| 4 | fiducial associated-W charge | Charge-separated fiducial rate. |
| 5–8 | top/antitop $p_{\mathrm{T}}, y$ | Truth top and antitop after QCD-radiation assignment. |
| 9–15 | $t\bar{t}$ mass, $p_{\mathrm{T}}$, $y$, $\Delta y$, $\Delta\|y\|$, $\Delta\phi$, $\Delta R$ | Standard pair-production spectra and asymmetry inputs. |
| 16–19 | scattering cosine, speed, $\rho_{tt}$, $p_{\mathrm{T}}$ asymmetry | $\cos\theta_t^*$, $\|\vec{p}_t^{\,*}\|/E_t^*$, $2m_t^{\text{ref}}/m_{tt}$, and normalized top-$p_{\mathrm{T}}$ difference. |
| 20–22 | associated-W $p_{\mathrm{T}}, y, m$ | Exact muon–neutrino W candidate. |
| 23–25 | $t\bar{t}W$ mass, $p_{\mathrm{T}}, y$ | Full heavy associated-production system. |
| 26 | $t\bar{t}W$ threshold $\rho$ | $(2m_t^{\text{ref}} + m_W^{\text{ref}})/m_{ttW}$. |
| 27 | heavy scalar $p_{\mathrm{T}}$ | $p_{\mathrm{T}}(t) + p_{\mathrm{T}}(\bar{t}) + p_{\mathrm{T}}(W_{\text{assoc}})$. |
| 28–31 | W–top, W–antitop, W–$t\bar{t}$ $\Delta R$; W–$t\bar{t}$ $\Delta\phi$ | Associated-W recoil geometry. |
| 32–34 | $q_W y_W$, $q_W y_{ttW}$, $q_W \Delta\|y_t\|$ | Charge-signed longitudinal and top-asymmetry projections. |

## 8.2 Charged leptons: histograms 35–70

| IDs | Histograms | Definition and purpose |
|---|---|---|
| 35–43 | $e^+, e^-, \mu$ $p_{\mathrm{T}}, \eta, y$ | Individual charged leptons. |
| 44–46 | ordered lepton $p_{\mathrm{T}}$ | Leading, subleading, and third lepton. |
| 47–50 | trilepton $m, p_{\mathrm{T}}, y, H_T^\ell$ | Global visible leptonic system. |
| 51–55 | OSSF $ee$ $m, p_{\mathrm{T}}, \Delta\phi, \|\Delta\eta\|, \Delta R$ | Top-decay dilepton system and Z-veto variables. |
| 56–60 | same-sign $m, p_{\mathrm{T}}, \|\Delta\phi\|, \|\Delta\eta\|, \Delta R$ | Signature pair; includes both ATLAS angular observables. |
| 61–64 | opposite-sign $e\mu$ $m, \Delta\phi, \|\Delta\eta\|, \Delta R$ | Complementary pair geometry. |
| 65–68 | min/max pair mass and min/max pair $\Delta R$ | Permutation-insensitive three-lepton topology. |
| 69–70 | $q_W \eta_\mu, q_W y_\mu$ | Charge-folded associated-lepton spectra. |

### 8.3 W/top decays and neutrinos: histograms 71–110

| IDs | Histograms | Definition and purpose |
|---|---|---|
| 71–79 | three W candidates, each $p_{\mathrm{T}}, y, m$ | Top-side, antitop-side, and associated W. |
| 80–82 | three-W $m, p_{\mathrm{T}}, y$ | Sum of all three W candidates. |
| 83–85 | correct $e^+b$, $e^-\bar{b}$, and maximum mass | Truth decay endpoints using bottom partons. |
| 86–87 | electron energies in parent-top frames | Charged-lepton decay spectra. |
| 88–90 | three W helicity cosines | Top, antitop, and associated-W polarization. |
| 91 | $q_W \cos\theta^*_{W,\mathrm{assoc}}$ | Common charge convention for W polarization. |
| 92–93 | top-side W energies in parent frames | Top-decay dynamics. |
| 94 | muon energy in associated-W frame | Off-shell-W and polarization sensitivity. |
| 95–100 | three neutrino $p_{\mathrm{T}}, \eta$ | Exact truth neutrinos by decay chain. |
| 101–103 | trineutrino $p_{\mathrm{T}}, m, y$ | Full invisible system. |
| 104 | $p_{\mathrm{T}}^{\mathrm{miss}}$ | Observable total missing transverse momentum. |
| 105–106 | min/max $\Delta\phi(p_{\mathrm{T}}^{\mathrm{miss}}, \ell)$ | Missing-momentum alignment. |
| 107 | $m_T(3\ell, p_{\mathrm{T}}^{\mathrm{miss}})$ | Trilepton cluster transverse mass. |
| 108–109 | $m_T(\ell_{\mathrm{lead}}, p_{\mathrm{T}}^{\mathrm{miss}})$, $m_T(\mu, p_{\mathrm{T}}^{\mathrm{miss}})$ | Measurable transverse masses. |
| 110 | missing-$p_{\mathrm{T}}$ closure | $\lvert \vec{p_{\mathrm{T}}^{\mathrm{miss}}} - \sum_\nu \vec{p_{\mathrm{T}}}_\nu \rvert$. |

### 8.4 Jets and fiducial observables: histograms 111–178

| IDs | Histograms | Definition and purpose |
|---|---|---|
| 111–116 | b/$\bar{b}$ $p_{\mathrm{T}}$, ordered b $p_{\mathrm{T}}$, individual $\eta$ | Ideal partonic b-tag kinematics. |
| 117–120 | bb $m, p_{\mathrm{T}}, \Delta\phi, \Delta R$ | Two-b topology; merged bottoms have zero separation. |
| 121–122 | $N_j(p_{\mathrm{T}} > 25)$, $N_j(p_{\mathrm{T}} > 40)$ | Central jet multiplicities. |
| 123–126 | leading/subleading jet $p_{\mathrm{T}}, \eta$ | Ordered accepted jets. |
| 127–130 | extra-jet multiplicity, leading-extra $p_{\mathrm{T}}, y$, $H_T^{\mathrm{extra}}$ | Radiation beyond the b jets. |
| 131–134 | $H_T^j, H_T^\ell, H_T^{\mathrm{vis}}, S_T$ | Event activity; first two are ATLAS unfolded variables. |
| 135–136 | visible $3\ell bb$ mass and $p_{\mathrm{T}}$ | Global reconstructible visible system. |
| 137 | $\Delta R_{\ell b,\mathrm{lead}}$ | Leading lepton to leading accepted b jet; ATLAS variable. |
| 138 | minimum lepton–b $\Delta R$ | Closest of six lepton–b combinations. |
| 139–140 | fiducial rate and charge | Baseline accepted cross sections. |
| 141–143 | fiducial ordered lepton $p_{\mathrm{T}}$ | Three accepted leptons. |
| 144–146 | fiducial leading/trailing electron and muon $\eta$ | Detector-acceptance shapes; electron ordering is by $p_{\mathrm{T}}$. |
| 147–150 | fiducial trilepton $m, p_{\mathrm{T}}, y, H_T^\ell$ | Accepted trilepton system. |
| 151–153 | fiducial dielectron $m, \Delta\phi, \Delta R$ | Defined for opposite- and same-sign dielectron events; the Z veto acts only on OSSF pairs. |
| 154–157 | fiducial same-sign $m, \lvert\Delta\phi\rvert, \lvert\Delta\eta\rvert, \Delta R$ | Signature pair and ATLAS angles. |
| 158–160 | fiducial leading opposite-sign pair $m, \Delta\phi, \Delta R$ | The unique opposite-charge lepton paired with the leading same-sign lepton. |
| 161–166 | $p_{\mathrm{T}}^{\mathrm{miss}}$, min/max $\Delta\phi$, leading/muon $m_T$, cluster $m_T$ | Neutrino-blind transverse observables. |
| 167–170 | ordered b $p_{\mathrm{T}}$, bb mass and $\Delta R$ | Visible heavy-flavour system. |
| 171–175 | jet/extra-jet multiplicity, $H_T^j, H_T^{\mathrm{vis}}, S_T$ | Accepted activity. |
| 176 | fiducial $\Delta R_{\ell b,\mathrm{lead}}$ | Measurement-facing ATLAS variable. |
| 177 | fiducial $m_{\mathrm{T2}}^{ee}$ | Uses total $p_{\mathrm{T}}^{\mathrm{miss}}$, including associated neutrino. |

| IDs | Histograms | Definition and purpose |
|---|---|---|
| 178 | accepted b-jet multiplicity | One-b fiducial versus two-b reconstruction tiers. |

### 8.5 Reconstruction and spin: histograms 179–278

| IDs | Histograms | Definition and purpose |
|---|---|---|
| 179 | reconstructed rate | Successful solutions within the two-b tier. |
| 180–188 | reconstructed three W candidates, each $p_{\mathrm{T}}, y, m$ | Mass-constraint-fit W bosons. |
| 189–194 | reconstructed top/antitop $p_{\mathrm{T}}, y, m$ | Both same-sign origins and both b-jet assignments tested. |
| 195–200 | reconstructed $t\bar{t}$, $t\bar{t}W$, each $m, p_{\mathrm{T}}, y$ | Full fitted heavy systems. |
| 201–206 | reconstructed neutrino $p_{\mathrm{T}}, p_z$ | Three fitted invisible components. |
| 207 | reconstructed trineutrino mass | Fitted invisible-system invariant mass. |
| 208–209 | fit score and b pairing | Minimum score; pairing 1 couples the positive/negative top leptons to b/$\bar{b}$, pairing 2 swaps the jets. |
| 210–212 | reconstructed W helicity cosines | Polarization after neutrino reconstruction. |
| 213 | reconstructed $q_W \cos\theta^*_{W,\mathrm{assoc}}$ | Charge-folded associated-W polarization. |
| 214 | reconstructed $t\bar{t}W$ threshold $\rho$ | Fitted heavy-threshold shape. |
| 215–220 | truth $c_k^+, c_r^+, c_n^+, c_k^-, c_r^-, c_n^-$ | Complete single-spin angles. |
| 221–229 | truth $q_{ab}$, $a, b \in \{k, r, n\}$ | All nine entries of the full $3 \times 3$ correlation matrix. |
| 230–240 | opening/$\Delta\phi$, symmetric and antisymmetric moments, triple product, diagonal estimator, signed sine | Derived spin and CP-sensitive projections. |
| 241–266 | reconstructed version of 215–240 | Same definitions using fitted tops. |
| 267–272 | truth associated-spin suite | Beam cosines, same-sign opening, raw/signed triple product, and associated-lepton–top contrast. |
| 273–278 | reconstructed associated-spin suite | Same six quantities in reconstructed frames. |

### 8.6 Diagnostics: histograms 279–286

| ID | Histogram | Meaning |
|---|---|---|
| 279 | truth top-assignment score | Minimum $\chi^2_{\mathrm{truth}}$. |
| 280 | assigned-radiation count | Light QCD partons assigned to either top decay. |
| 281 | truth top-mass residual | Larger top/antitop reference-mass deviation. |
| 282 | missing-$p_{\mathrm{T}}$ closure residual | Numerical event-record consistency. |
| 283 | reconstruction score | Minimum $\chi^2_{\mathrm{reco}}$, repeated for monitoring. |
| 284 | reconstructed W-mass residual | Largest of three W reference-mass deviations. |
| 285 | reconstructed top-mass residual | Larger fitted top-mass deviation. |
| 286 | reconstruction rate | One-bin normalization for successful fits. |

## 9 IR safety and fixed-order use

The analysis never branches on `ibody`. The same code sees Born, real, soft, collinear, and soft-collinear projected points in additive and multiplicative decay-chain calculations. All measurement inputs are infrared-safe functions of final momenta: anti-$k_T$ jets, momentum sums, ordered objects away from measure-zero ties, and ordinary acceptance cuts. In particular, collinear $\xi = 0, y = 1$ projections are experimentally indistinguishable and are evaluated identically. Histogram boundaries, jet-assignment boundaries, and mass-fit solution crossings move an event only on measure-zero surfaces, as in any binned IR-safe analysis.

For a meaningful charge ratio, both $W^+$ and $W^-$ processes must be included with identical settings. If only one charge is generated, the charge histograms remain valid bookkeeping but do not define the hadronic asymmetry.

# 10 Parton-level limitations

This module is more realistic than a no-cut truth analysis but is not a detector simulation. Leptons are bare QCD-level particles rather than photon-dressed objects; b tags use bottom partons rather than ghost-associated b hadrons; there is no detector efficiency, resolution, pile-up, fake lepton, or charge-misidentification model. The $2e1\mu$ flavour channel is one component of an experimental inclusive $e/\mu$ result. Reconstruction solutions are prescription-dependent, and truth residuals are validation plots rather than measurable observables.

The fiducial cuts are a reproducible theory region, not a claim of bitwise identity with an ATLAS reconstruction. Direct data comparisons require a matched particle-level definition, branching-ratio and flavour factors, electroweak/QED treatment, and non-perturbative corrections.

# 11 Implementation and selection

Select the module with

```
FO_ANALYSIS_FORMAT=HwU
FO_ANALYSE=analysis_HwU_pp_ttxw_trilepton.o
```

The fNLO makefile automatically adds `analysis_HwU_pp_ttxw_trilepton_bridge.o`. The bridge exports the ordinary `analysis_fill` ABI and variable-length `analysis_fill_multiplicative` ABI. FastJet is linked through the standard fNLO wrapper.

A representative $W^+$ process is

```
p p > t t~ w+ [QCD],
 (t  > w+ b  QED=1 [QCD], w+ > e+ ve),
 (t~ > w- b~ QED=1 [QCD], w- > e- ve~),
 w+ > mu+ vm
```

with the charge-conjugate associated-W decay for $W^-$. Exact syntax can depend on the model and decay-chain generation mode; object mapping in the analysis does not depend on written order. The visible and reconstructed histograms also accept, for example, a $W^+$ configuration with $e^+e^+\mu^-$. The flavour-specific truth histograms remain empty for such an alternative origin assignment, by construction.

# 12 References


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